

ALLAN PENG

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WORK EXPERIENCE

Databricks

Sr. Software Engineer, Cluster Team

2021 -2025

- Engineer on the Cluster Manager team, the compute control plane underlying all of Databricks, which launches 10s of millions of VMs across 50+ cloud regions daily and supporting \$4B+ in annual revenue.
- Owned the team's largest bottleneck, database instability, partnering closely with the storage team to develop deep expertise in storage internals and query optimization. Drove a multi-quarter effort that reduced weekly region-wide outages to effectively zero, directly unblocking the Zero Downtime Cluster Manager project and freeing investment to shift to other strategic priorities.
- Resolved production incidents spanning multiple systems: schema migration tooling causing prolonged database lockups, feature flag infrastructure causing regional degradation, character encoding issues in Kafka/CDC pipelines causing data corruption
- Led a series of stability projects: made query performance bounded and deterministic, reduced PMAX query time from O(minutes) to under 1 second, optimized connection pools, migrated our databases to MySQL8.0 and to TiDB
- Prototyped a dataplane worker health monitoring system for detecting and remediating unhealthy worker nodes; project was subsequently completed and shipped by the team.
- Recognized as a go-to technical resource across engineering; gave multiple org-wide talks on system internals and production incident analysis; onboarded the Amsterdam engineering team through in-person knowledge transfer and ongoing consultation.

Mod9 Technologies

Software Engineer

2018-2021

- Led development of the company's core real-time speech recognition server in C++/Boost as technical lead
- Built parallelized recognizer capable of transcribing audio up to 400× faster than real time
- Implemented new speaker diarization (segmentation) system, using agglomerative clustering and HMM-GMM models improving diarization error rate from 15% to 4%.
- Trained HMM-DNN models to transcribe English and Spanish using Kaldi.

SKILLS

Languages: Java, C, C++/Boost, Python, Go, Scala, SQL

Infrastructure: Distributed systems, Kafka, MySQL, TiDB, Docker, Kubernetes, Kaldi

EDUCATION

University of California, Berkeley

B.A. Double Major: Applied Math & Computer Science; Regents & Chancellor's Scholarship

London School of Economics

International Relations & Cyberlaw